

"If JavaScript is a programming language, how does it change something on an HTML webpage?"

The answer is :



2. What is DOM?

DOM = Document Object Model

The DOM is a programmatic representation of an HTML document created by the browser.

When the browser loads:

```
<!DOCTYPE html>
<html>
  <body>
    <h1>Hello</h1>
    <p>Welcome</p>
  </body>
</html>
```

the browser creates a structure that JavaScript can access and manipulate.

Think of it as:
HTML Source Code

↓
Browser
↓
DOM

↓
JavaScript can interact with it

Simple definition :

The DOM is the browser's representation of an HTML page as a tree of objects/nodes that JavaScript can work with.

3. HTML VS DOM ?

The distinction is important “DOM is HTML” – not exactly

HTML → input

DOM → browser's structured representation

HTML

HTML is the markup/code we write.

```
<h1>Hello</h1>
```

DOM :

The browser reads that HTML and creates an object-based structure.

Document



html



body



h1



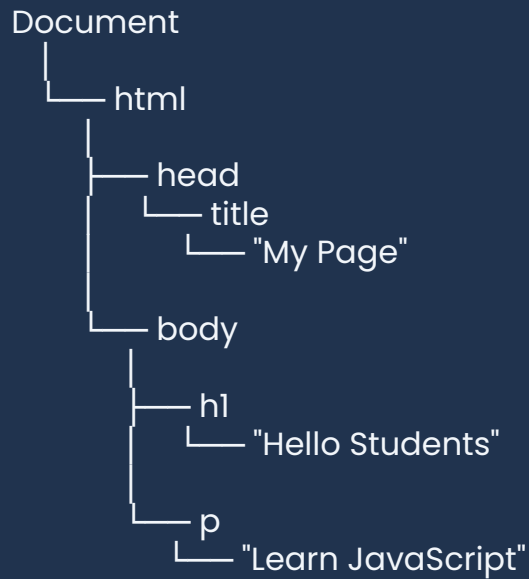
"Hello"

4. What is DOM Tree ?

The DOM represents the webpage as a tree structure

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Page</title>
  </head>

  <body>
    <h1>Hello Students</h1>
    <p>Learn JavaScript</p>
  </body>
</html>
```



5. Why a Tree ?

Because elements have relationships

```
<body>  
  <h1>Hello</h1>  
  <p>Welcome</p>  
</body>
```

Body -
├── h1 -
└── p

body is the parent.
h1 and p are children.

```
body  
├── h1  
├── p  
└── button
```

body → parent
h1 → child of body
p → child of body
button → child of body
h1, p, button → siblings

This becomes useful later when we work with DOM traversal.

6. What is a document ?

Javascript provides a special object called document
document

It represents the webpage/document currently loaded in the browser

For example:

```
console.log(document);
```

run this in the browser console.

They will see information representing the current webpage.

Conceptually:

Browser



So when we write

document

We are essentially saying

“Give me access to this webpage / document”

7. What is a Element ?

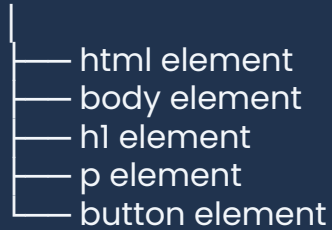
`<h1>Hello</h1>`

`<p>Welcome</p>`

`<button>Click</button>` - h1 , p and button are elements

Conceptually

DOM



8. What is a Node ?

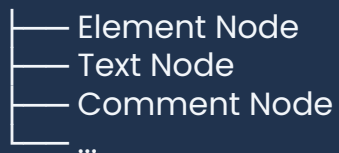
Node is a broader concept than element

For eg : `<h1>Hello</h1>`

H1 - Element Node , "Hello" - Text node

So

Node



9. Element Vs node ?

<p>Hello Students</p>

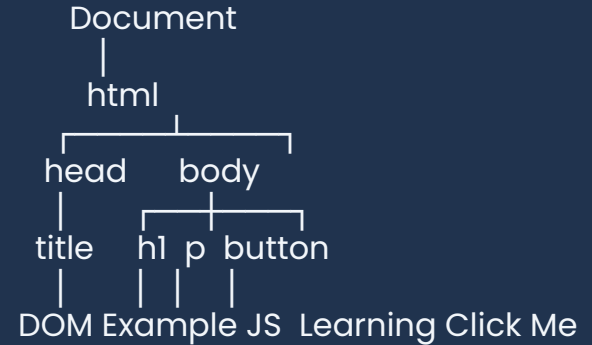
<p> → Element Node
"Hello Students" → Text Node

Therefore:

Every element is a node, but not every node is an element.

10. A simple DOM Example

```
<!DOCTYPE html>
<html>
<head>
  <title>DOM Example</title>
</head>
<body>
  <h1>JavaScript</h1>
  <p>Learning DOM</p>
  <button>Click Me</button>
</body>
</html>
```



11. Look at Javascript + DOM

```
console.log(document);  
console.log(document.body);  
console.log(document.body.children);
```

Summary :

